

v1.12.9 Deep Audit — what the green build was hiding

"test_routes 87/87 PASS" was testing NONE of the v1.12 work

FOUND (real)

8 new routes: ZERO coverage
68 of 133 API routes untested
decoders UNREACHABLE (no UI)
/reference clashed w/ /api/reference
verify_ui gate = hardcoded list
green build, 51% of API never hit

FIXED (structural)

curated cases 87 → 106
+ error paths (400 / null / reasons)
+ **blanket registry.GET sweep**
every route hit bare → no 5xx
new routes auto-covered forever
/reference → /decode (no clash)

REACHABLE + HARDENED

/decode page + palette command
paste any code → auto-detect kind
NSN / SMR / CAGE / MS-MIL-SAE
nulls shown honestly, not hidden
24,000 fuzz cases: 0 crashes
R13 invariants held under fuzz

R13 invariants asserted under 24k hostile inputs

curated flag ↔ named item agree · NIIN=9 / FSC=4 · every name is None-or-str (NEVER fabricated)
CAGE valid XOR reasons · harness nets never singleton · MAC rows always name a real function

The point of the audit: a passing suite that never runs the code is worse than no suite —
it buys false confidence. The sweep makes that failure mode structurally impossible.

additive (R1) · verification logic strengthened, never weakened · VERSION 1.12.9
no duplicate routes · no missing page files · all 42 palette destinations resolve